Premier University, Chittagong

Department of Computer Science and Engineering

Final Project Report

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Chapter 1
Introduction

1.1 Introduction

The Hostel Management System is a web-based software project developed to manage hostel meal, billing, complaint, notice, and administrative activities from a single platform. The project separates user responsibilities into student, manager, and administrator roles so that each user sees only the features needed for their work.

1.2 Motivation

Manual hostel management often depends on paper records, message groups, and verbal communication. This can cause incorrect meal counts, delayed notices, billing confusion, and weak complaint tracking. A digital system can reduce these problems by storing information in a structured and traceable way.

1.3 Objectives

1. To provide a digital platform for hostel service management.
2. To support role-based access for students, managers, and administrators.
3. To make meal planning, attendance, billing, notices, and complaints easier to monitor.

Chapter 2
Software Requirements Specification

2.1 Functional Requirements

- Users shall be able to register and log in securely.
- Students shall be able to view menus, submit meal choices, check bills, and submit complaints.
- Managers shall be able to manage menus, attendance, and inventory.
- Administrators shall be able to manage users, billing, notices, analytics, and settings.

2.2 Non-functional Requirements

- The system should be responsive on desktop and mobile browsers.
- Passwords should be stored securely using hashing.
- The backend should validate user input and protect private routes.
- The codebase should be modular so that future features can be added easily.

Chapter 3
Methodology

3.1 Architecture

The project follows a client-server architecture. The frontend communicates with a backend API, and the backend stores persistent data in a database. Authentication middleware protects role-specific routes.

Figure 3.1: Demo system architecture

Chapter 4
System Implementation Overview

4.1 Features of the System

Module	Implemented Capability
Authentication	Login, registration, profile access, and protected routes.
Meal Management	Menu viewing, meal selection, and attendance support.
Billing	Monthly bill generation and student bill tracking.
Complaints	Student complaint submission and administrative review.
Notices	Administrator notice publication and user notice viewing.

Chapter 5
Risk Analysis

5.1 Assessing the Risk of Project

Risk	Probability	Impact	Assessment
Incorrect meal data	Medium	High	Wrong meal counts can affect food planning and billing.
Unauthorized access	Low	High	Protected modules must remain role-restricted.
Billing errors	Medium	High	Financial records require careful validation.

Table 5.1: Demo project risk assessment

Chapter 6
Social and Security Impact

6.1 Social Impact

The system can improve transparency between hostel residents and management by making meal, notice, complaint, and billing information easier to access.

6.2 Security and Reliability

The project should use secure authentication, password hashing, input validation, and role-based access control. Reliability can be improved through testing, logging, and modular design.

Chapter 7
Conclusion

The Hostel Management System demonstrates how software engineering practices can solve real hostel administration problems. The demo report shows the expected structure for a full academic project report.